Taner Cokyasar

Assistant Professor, Engineering Technology and Industrial Distribution (ETID)
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At a Glance

Operations researcher building large-scale optimization models and algorithms for transportation, logistics, and energy systems, including EV charging and fleet electrification, transit, freight, drone delivery, and shared mobility. Seven years at Argonne National Laboratory turning these methods into national- and metro-scale studies; now directing the ATLAS Lab at Texas A&M University.

2020 — Ph.D., Industrial Engineering, Univ. of Tennessee
→ **2019–2026** — Intern → Comp. Transp. Engineer, Argonne National Lab.
→ **2021–2022** — Asst. Prof., Industrial Eng., Tarsus University
→ **2026–present** — Asst. Prof., ETID, Texas A&M University

$605K	**Area Editor**	**20**	**15**	**524**
funding as PI	*The Engineering Economist*	journal articles	under review	citations
4 DOE awards	since 2025	+ 15 conf. papers	6 journal + 9 conf.	h-index 13, i10 14
65	**17**	**14**	**52**	**78**
unique co-authors	institutions	students mentored	talks, posters, panels	peer reviews
across all papers	across all papers	2 Ph.D. co-advised	since 2018	for 16 venues

Research & Funding
- PI on four DOE awards through Argonne ($605K)
- Journals include *Transportation Research Part B* (2) and *Part C* (5), *EJOR* (2), *IISE Transactions*, *Computers & Operations Research*

Mentoring & Teaching
- Ph.D. chair (Texas A&M ISEN); co-advised 2 Ph.D. dissertations (UIC); mentees from 8 universities, now at Uber, NetJets, Starbucks, Wells Fargo, and Plasman
- IDIS 450 *Analytics for Distribution Operations*, 97 students (Fall 2026); 6 earlier course offerings in the U.S. and Türkiye

Leadership & Service
- President, IISE Engineering Economy Division (2025–2027); Area Editor, *The Engineering Economist* (2025–present)
- Track chair, IISE Annual Meeting (2023–2026), EUSPN, and Mobi-SPC; session chair, IISE and INFORMS 2026; invited panelist, IISE 2026; 2 ETID faculty search committees

Selected Honors
- Impact Argonne Award (2024) • IISE Best Track Paper Award (2018)
- E.J. Sierleja Memorial Fellowship, IISE (2019–20) • Extraordinary Professional Promise Award, Univ. of Tennessee (2018)

Research Interests

My research develops large-scale optimization models and algorithms that make transportation and logistics systems more efficient, affordable, and resilient. I combine mathematical programming, decomposition and high-performance computing with agent-based simulation (POLARIS) to answer planning questions at metropolitan and national scale: where to place EV and electric-bus chargers, how to schedule transit and delivery fleets and transition them to new vehicle technologies, how drones and trucks should share last-mile delivery, and how shared and autonomous mobility and freight supplier choices reshape travel, energy use, and cost.

Education

***The University of Tennessee*, Knoxville, TN, U.S.A.**

01/2017 – 05/2020 • Ph.D. in Industrial Engineering
Dissertation: *Economic Feasibility of Adopting Additive Manufacturing for Low-volume Demand*
Committee: *Mingzhou Jin (advisor), Sean Willems, Xueping Li, James Ostrowski*

08/2015 – 12/2016 • M.S. in Industrial Engineering
Thesis: *A Mixed Integer Linear Programming Approach for Developing Salary Administration Systems*
Committee: *Alberto Garcia-Diaz (advisor), John Kobza, Rapinder Sawhney*

***The University of Georgia*, Athens, GA, U.S.A.**

08/2014 – 08/2015 • Non-degree Intensive English Program at the English Language Institute

***Nevsehir Haci Bektas Veli University*, Nevsehir, Türkiye**

09/2008 – 06/2012 • B.S. in Business Administration

Academic and Research Appointments

***Texas A&M University*, College Station, TX, U.S.A.**

08/2026 – ····· • Assistant Professor - Engineering Technology and Industrial Distribution Department
08/2026 – ····· • Director - Advanced Transportation Logistics & AI for Resilient Systems (ATLAS) Lab

***Argonne National Laboratory*, Lemont, IL, U.S.A.**

07/2024 – 08/2026 • Computational Transportation Engineer
05/2021 – 07/2024 • Research Consultant
05/2020 – 05/2021 • Postdoctoral Appointee Transportation System Modeler
01/2020 – 05/2020 • Visiting Student (Consultant)
05/2019 – 08/2019 • W. J. Cody Associate Intern

***Tarsus University*, Mersin, Türkiye**

11/2021 – 09/2022 • Assistant Professor - Industrial Engineering Department
03/2021 – 11/2021 • Lecturer - Industrial Engineering Department

***Toros University*, Mersin, Türkiye**

03/2021 – 11/2021 • Lecturer - Industrial Engineering Department

***The University of Tennessee*, Knoxville, TN, U.S.A.**

03/2017 – 05/2020 • Graduate Research/Teaching Assistant

***Oak Ridge National Laboratory*, Oak Ridge, TN, U.S.A.**

05/2018 – 08/2018 • Advanced Short-Term Research Opportunity (ASTRO) Intern

Research Funding

Active – Texas A&M University

08/2026 – 08/2027 • *Principal Investigator* – Developing Optimization Models and Algorithms to Study Large-Scale Transportation Systems, U.S. Department of Energy through Argonne National Laboratory ($154,622)

Pending

Submitted 2026 • *Principal Investigator* – Vehicle Miles Travelled Impacts of Land Use Development Patterns, Minnesota Department of Transportation ($955,670)

Submitted 2026 • *Principal Investigator, Texas A&M subaward* – Evidence-Based Supplier Sufficiency and Qualification Readiness for ISAM Interfaces, NASA MUREP M-STAR through Texas State University ($400,000)

Submitted 2026 • *Principal Investigator* – Evaluating Use Cases of Cargo E-Bikes in Minnesota, Minnesota Department of Transportation ($375,190)

Pre-proposals and White Papers

2026 • *Sole PI* – Polyhedral and Decomposition Methods for Resilient Replenishment-Constrained Network Optimization, AFOSR Young Investigator Program, white paper

2026 • *Lead author (2), co-author (1)* – Texas Department of Transportation FY2028 Research Program problem statements: connected-vehicle data and metropolitan simulation for congestion and signal investment; MD/HD vehicle technologies and Texas freight costs; phased advanced air mobility deployment in Texas

Completed – Argonne National Laboratory, Lemont, IL, U.S.A.

05/2021 – 07/2024 • *Principal Investigator* – Three consecutive awards from the U.S. Department of Energy through Argonne National Laboratory ($450,000 total):

05/2023 – 07/2024 • Optimization for Transit, Freight, and Urban Mobility Problems ($200,000)

05/2022 – 05/2023 • Optimization for Transit and Freight ($130,000)

05/2021 – 05/2022 • Development and Implementation of Optimization Models to Support Transportation and Mobility ($120,000)

2020 – 2021 • *Postdoctoral Researcher* – Department of Energy (DOE) High Performance Computing for Mobility (HPC4Mobility): Chicago Transit Authority Transit Network Efficiency and the Changing Mobility Landscape ($25,000)

2020 – 2021 • *Postdoctoral Researcher* – Department of Energy (DOE) Systems & Modeling for Accelerated Research in Transportation (SMART) Mobility 2.0 ($50,000)

01/2020 – 05/2020 • *Graduate Researcher* – Enhancing POLARIS Agent-Based Framework ($60,000)

05/2019 – 08/2019 • *Graduate Researcher* – Optimal Assignment for the Single-Household Shared Autonomous Vehicle Problem ($20,000)

Completed – Oak Ridge National Laboratory, Oak Ridge, TN, U.S.A.

05/2018 – 08/2018 • *Graduate Researcher* – Optimization for Smart Manufacturing Technology Adoption ($15,000)

Publications and Conferences

Journal Articles (*Corresponding Author; underline: mentored student)

20 Hosseini Dolatabadi, S. H., Bhuiyan, T. H., Kaleem, W., Subramanyam, A., & **Cokyasar, T.** (2026). A branch-and-cut algorithm for routing a heterogeneous drone-ground vehicle fleet to deliver time-sensitive products. *Transportation Research Part C: Emerging Technologies*, *183*. https://doi.org/10.1016/j.trc.2025.105454

19 Ismael, A., & **Cokyasar, T.** (2026). Modeling and calibration of the supplier selection problem in freight agent–based simulations. *Transportation Research Record*. https://doi.org/10.1177/03611981261455027

18. Kaleem, W., **Cokyasar**[*], **T.**, Larson, J., Verbas, O., Bhuiyan, T. H., & Subramanyam, A. (2026). Extreme-scale EV charging infrastructure planning for last-mile delivery using high-performance parallel computing [In Press]. *Transportation Research Part B: Methodological*, *205*. https://doi.org/10.1016/j.trb.2026.103403
17. Ng, T., Max, Mahmassani, H., Tong, D., Verbas, O., & **Cokyasar**, **T.** (2026). Joint optimization of multimodal transit frequency and shared autonomous vehicle fleet size with hybrid metaheuristic and nonlinear programming. *Transportation Research Part C: Emerging Technologies*, *185*, 105568. https://doi.org/10.1016/j.trc.2026.105568
16. Verbas, O., **Cokyasar**, **T.**, Joyce-Johnson, S., Wainwright, S., Coates, M., Rousseau, A., Aloisi, J., Stewart, A., & Auld, J. (2026). Impact of transit on mobility, equity, and economy in the chicago metropolitan region [Accepted]. *Public Transport*. https://arxiv.org/pdf/2409.04568
15. Bazarnovi, S., **Cokyasar**[*], **T.**, Verbas, O., & Mohammadian, A. K. (2025). Problem of locating and allocating charging equipment for battery electric buses under stochastic charging demand. *European Journal of Operational Research*. https://doi.org/10.1016/j.ejor.2025.07.064
14. Paul, J., Gurumurthy, K. M., **Cokyasar**, **T.**, Su, H., Khan, N., Auld, J., & Jia, Y. (2025). Proactive assignment strategy with human choice models for boosting pooled rideshare service. *IEEE Transactions on Intelligent Transportation Systems*, *26*, 2351–2365. https://doi.org/10.1109/TITS.2024.3515074
13. Davatgari, A., **Cokyasar**[*], **T.**, Subramanyam, A., Larson, J., & Mohammadian, A. K. (2024). Electric vehicle supply equipment location and capacity allocation problem. *European Journal of Operational Research*, *317*, 953–966. https://doi.org/10.1016/j.ejor.2024.04.022
12. Davatgari, A., **Cokyasar**, **T.**, Verbas, O., & Mohammadian, A. K. (2024). Heuristic solutions to the single depot electric vehicle scheduling problem with next day operability constraints. *Transportation Research Part C: Emerging Technologies*, *163*, 104656. https://doi.org/10.1016/j.trc.2024.104656
11. Dean, M. D., Gurumurthy, K. M., Zhou, Z., Verbas, O., **Cokyasar**, **T.**, & Kockelman, K. (2024). An integrated transportation-power system model for a decarbonizing world. *Transportation Planning and Technology*, *47*, 1277–1313. https://doi.org/10.1080/03081060.2024.2348721
10. Ng, M. T. M., Mahmassani, H. S., Verbas, O., **Cokyasar**, **T.**, & Engelhardt, R. (2024). Redesigning large-scale multimodal transit networks with shared autonomous mobility services. *Transportation Research Part C: Emerging Technologies*, *168*, 104575. https://doi.org/10.1016/j.trc.2024.104575
9. **Cokyasar**[*], **T.**, Stinson, M., Sahin, O., Prabhakar, N., & Karbowski, D. (2023). Comparing regional energy consumption for direct drone and truck deliveries. *Transportation Research Record*, *2677*(2), 310–327. https://doi.org/10.1177/03611981221145137
8. **Cokyasar**[*], **T.**, de Souza, F., Auld, J., & Verbas, O. (2022). Dynamic ride-matching for large-scale transportation systems. *Transportation Research Record*, *2676*(3), 172–182. https://doi.org/10.1177/03611981211049422
7. **Cokyasar**, **T.**, & Jin, M. (2022). Additive manufacturing capacity allocation problem over a network. *IISE Transactions*, *55*(8), 807–820. https://doi.org/10.1080/24725854.2022.2120222
6. **Cokyasar**[*], **T.**, Subramanyam, A., Larson, J., Stinson, M., & Sahin, O. (2022). Time-constrained capacitated vehicle routing problem in urban e-commerce delivery. *Transportation Research Record*, *2677*(2), 190–203. https://doi.org/10.1177/03611981221124592
5. Subramanyam, A., **Cokyasar**, **T.**, Larson, J., & Stinson, M. (2022). Joint routing of conventional and range-extended electric vehicles in a large metropolitan network. *Transportation Research Part C: Emerging Technologies*, *144*, 103830. https://doi.org/10.1016/j.trc.2022.103830
4. **Cokyasar**[*], **T.** (2021). Optimization of battery swapping infrastructure for e-commerce drone delivery. *Computer Communications*, *168*, 146–154. https://doi.org/10.1016/j.comcom.2020.12.015
3. **Cokyasar**[*], **T.**, Dong, W., Jin, M., & Verbas, İ. Ö. (2021). Designing a drone delivery network with automated battery swapping machines. *Computers & Operations Research*, *129*, 105177. https://doi.org/10.1016/j.cor.2020.105177

2. **Cokyasar**, **T.**, & Larson, J. (2020). Optimal assignment for the single-household shared autonomous vehicle problem. *Transportation Research Part B: Methodological*, *141*, 98–115. https://doi.org/10.1016/j.trb.2020.09.003

1. **Cokyasar**, **T.**, Garcia-Diaz, A., & Jin, M. (2019). Optimization of size and timing of base salary increases. *The Engineering Economist*, *64*(2), 97–115. https://doi.org/10.1080/0013791X.2018.1528408

In Progress and Submitted Manuscripts

16. Alam, M. R., Verbas, O., **Cokyasar**, **T.**, Bhidya, H., Altman, J., Beck, N., Auld, J., Veiga de Camargo, P., Cook, J., de Souza, F., Nair, G., Uhm, H., & Zill, J. (n.d.). Modeling of mobility and energy policies in an agent-based framework: Case studies for Chicago region in 2050. Submitted to the *106th Transportation Research Board Annual Meeting (Washington, DC, US, 01/10/2027 - 01/14/2027)*.

15. Bazarnovi, S., **Cokyasar**, **T.**, Verbas, O., & Mohammadian, A. (n.d.). Integrated optimization of scheduling and flexible charging in mixed electric-diesel urban transit bus systems. Submitted to the *Transportation Research Part E: Logistics and Transportation Review* [Under Review]. https://arxiv.org/pdf/2601.11751

14. Bhidya, H., **Cokyasar**, **T.**, & Verbas, O. (n.d.). Bus fleet electrification under capital cost and scheduling constraints: A five-agency case study. Submitted to the *106th Transportation Research Board Annual Meeting (Washington, DC, US, 01/10/2027 - 01/14/2027)*.

13. Bhidya, H., **Cokyasar***, **T.**, & Verbas, O. (n.d.). Modeling battery-electric bus fleet transition efficiency across five transit systems. Abstract submitted to the *2027 Modeling Mobility (MoMo) Conference (Tempe, AZ, US, 04/12/2027 - 04/15/2027)*.

12. **Cokyasar**, **T.** (n.d.). Intercity EV charging readiness in Türkiye: A countrywide scheduling and economic assessment. Submitted to the *Journal of Turkish Operations Management*. https://arxiv.org/pdf/2607.25143

11. **Cokyasar**, **T.** (n.d.). A national-scale ev charging scheduling framework: Optimal detour routing under infrastructure capacity constraints. Submitted to the *106th Transportation Research Board Annual Meeting (Washington, DC, US, 01/10/2027 - 01/14/2027)*. https://arxiv.org/pdf/2607.18453

10. Hajibagher Najjar, Z., **Cokyasar**, **T.**, Alam, M., & Mohammadian, A. K. (n.d.). Joint fleet sizing, planning, and electrification for last-mile delivery: A two-stage continuous approximation and optimization approach. Submitted to the *106th Transportation Research Board Annual Meeting (Washington, DC, US, 01/10/2027 - 01/14/2027)*.

9. Ismael, A., **Cokyasar**, **T.**, & Huang, Y. (n.d.). A national-scale supplier selection framework for freight agent-based simulation. Submitted to the *106th Transportation Research Board Annual Meeting (Washington, DC, US, 01/10/2027 - 01/14/2027)*.

8. Kaleem, W., **Cokyasar**, **T.**, Ismael, A., Sahin, O., & Uhm, H. S. (n.d.). Approximate dynamic programming for matching, scheduling, and pricing in truckload freight markets. Submitted to the *106th Transportation Research Board Annual Meeting (Washington, DC, US, 01/10/2027 - 01/14/2027)*.

7. Ng, T., Max, Raymer, M., Mahmassani, H., Verbas, O., & **Cokyasar**, **T.** (n.d.). Equity impacts of public transit network redesign with shared autonomous mobility services. Submitted to the *Public Transport* - [Under Review]. https://arxiv.org/abs/2501.01615

6. Paul, J., Su, H., Gurumurthy, K. M., **Cokyasar**, **T.**, Auld, J., & Jia, Y. (n.d.). A study of financial impacts of pooled rideshare based on assignment strategies. Submitted to the *Transportation Research Part A: Policy and Practice* [Under Review]. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5101649

5. Paul, J., Vankadaru, V., Gurumurthy, K. M., **Cokyasar**, **T.**, Auld, J., & Jia, Y. (n.d.). Strategic deployment of tncs for sustainable urban transportation: Insights from Austin. Submitted to the *IEEE Transactions on Intelligent Transportation Systems* - [Under Review].

4. Vakayil, A., de Souza, F., **Cokyasar**, **T.**, Gurumurthy, K. M., & Larson, J. (n.d.). Large-scale dynamic ridesharing with iterative assignment. https://arxiv.org/abs/2311.06160

3. Verbas, O., **Cokyasar**, **T.**, de Camargo, P. V., Gurumurthy, K. M., Zuniga-Garcia, N., & Auld, J. (n.d.). Modeling transit in a fully integrated agent-based framework: Methodology and large-scale application. Submitted to the *Public Transport* - [Under Review]. https://arxiv.org/pdf/2408.05176
2. Wu, Y., **Cokyasar***, **T.**, Jayashankar, N., Gurumurthy, K. M., & Larson, J. (n.d.). Multi-household shared autonomous vehicle problem with subscription tiers. Abstract submitted to the *2027 Modeling Mobility (MoMo) Conference (Tempe, AZ, US, 04/12/2027 - 04/15/2027)*.
1. Wu, Y., **Cokyasar**, **T.**, Jayashankar, N., Gurumurthy, K. M., & Larson, J. (n.d.). Multi-household shared autonomous vehicle problem with subscription tiers. Submitted to the *106th Transportation Research Board Annual Meeting (Washington, DC, US, 01/10/2027 - 01/14/2027)*.

Conference Papers

15. Acosta, J. G., Verbas, O., **Cokyasar**, **T.**, Auld, J., Altman, J., Beck, N., Alhajjar, M., de Camargo, P. V., Cook, J., & Derrible, S. (2026). Agent-based modeling of transportation energy shifts and trends: A case study of the Chicago region. In: *The 17th International Conference on Ambient Systems, Networks and Technologies (ANT), Procedia Computer Science, 280*, 598–605. https://doi.org/10.1016/j.procs.2026.04.076
14. Bazarnovi, S., **Cokyasar***, **T.**, Verbas, O., & Mohammadian, A. K. (2025). Optimizing charging infrastructure for battery electric buses: A stochastic location model with budget constraints. In: *The 16th International Conference on Ambient Systems, Networks and Technologies (ANT), Procedia Computer Science, 257*, 583–590.
13. **Cokyasar***, **T.**, Bazarnovi, S., Verbas, O., Ismael, A., & Mohammadian, A. K. (2025). Assessing the impact of route interlining via bus schedule optimization. In: *The 16th International Conference on Emerging Ubiquitous Systems and Pervasive Networks (EUSPN), Procedia Computer Science, 272*, 179–186. https://doi.org/10.1016/j.procs.2025.10.194
12. Davatgari, A., Mohammadi, M. Y., **Cokyasar**, **T.**, Mohammadian, A. K., & Auld, J. (2024). A label-correcting algorithm for constrained one-to-many k-shortest path problem with replenishment. In: *The 15th International Conference on Ambient Systems, Networks and Technologies (ANT), Procedia Computer Science, 238*, 369–376. https://doi.org/10.1016/j.procs.2024.06.037
11. Davatgari, A., Verbas, O., **Cokyasar**, **T.**, & Mohammadian, A. K. (2024). Optimization of electric bus scheduling and charger location. In: *The 15th International Conference on Ambient Systems, Networks and Technologies (ANT), Procedia Computer Science, 238*, 361–368. https://doi.org/10.1016/j.procs.2024.06.036
10. **Cokyasar***, **T.**, Davatgari, A., & Mohammadian, A. K. (2023). An optimization model for solving the route clustering problem. In: *The 14th International Conference on Ambient Systems, Networks and Technologies (ANT), Procedia Computer Science, 220*, 180–186. https://doi.org/10.1016/j.procs.2023.03.025
9. **Cokyasar***, **T.**, Verbas, O., & Auld, J. (2023). Electric vehicle scheduling problem with tour combinations. In: *The 14th International Conference on Ambient Systems, Networks and Technologies (ANT), Procedia Computer Science, 220*, 413–420. https://doi.org/10.1016/j.procs.2023.03.053
8. **Cokyasar***, **T.**, Verbas, O., Davatgari, A., & Mohammadian, A. K. (2023). Solving the electric vehicle scheduling problem at large-scale. In: *IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)*, 1134–1139. https://doi.org/10.1109/ITSC57777.2023.10422441
7. Paul, J., Gurumurthy, K. M., **Cokyasar**, **T.**, Su, H., Auld, J., & Jia, Y. (2023). Optimization of dynamic ride-sharing by considering user preference through discount and delay tolerance. In: *IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)*, 2770–2775. https://doi.org/10.1109/ITSC57777.2023.10422612
6. **Cokyasar***, **T.** (2021). Delivery drone route planning over a battery swapping network. In: *The 12th International Conference on Ambient Systems, Networks and Technologies (ANT), Procedia Computer Science, 184*, 10–16. https://doi.org/10.1016/j.procs.2021.03.013
5. **Cokyasar***, **T.**, Auld, J., Javanmardi, M., Verbas, O., & de Souza, F. (2020). Analyzing energy and mobility impacts of privately-owned autonomous vehicles. In: *IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)*, 1–6. https://doi.org/10.1109/ITSC45102.2020.9294218

4. **Cokyasar**, **T.**, Dong, W., & Jin, M. (2019). Network optimization for hybrid last-mile delivery with trucks and drones. In: *IISE Annual Conference Proceedings*, 1250–1255, ISBN: 978-0-9837624-8-5.
3. **Cokyasar**, **T.**, Kose, H., Dong, W., Li, X., Jin, M., & Li, W. (2018). An agent-based simulation model for optimization of spare parts sourcing with additive manufacturing. In: *IISE Annual Conference Proceedings*, 982–987, ISBN: 978-1- 5108-6935-6.
2. Hassler, M., **Cokyasar**, **T.**, Jin, M., & Tang, R. (2018). Optimization of maintenance and infrastructure projects based on risk factors. In: *IISE Annual Conference Proceedings*, 624–629, ISBN: 978-1- 5108-6935-6.
1. **Cokyasar**, **T.**, Garcia-Arcaya, C., & Garcia-Diaz, A. (2017). A mixed integer linear programming MATLAB-based methodology for the development of a salary administration guide. In: *IISE Annual Conference Proceedings*, 506–511, ISBN: 978-0-9837624-6-1.

Invited Talks and Panels

6. Bazarnovi, S., **Cokyasar**, **T.**, Verbas, O., & Mohammadian, A. K. (2026). Integrated optimization of scheduling and flexible charging in mixed electric-diesel urban transit bus systems. *INFORMS Annual Meeting*. Invited abstract will be presented by Cokyasar, T. in San Francisco, CA, November 1-4.
5. **Cokyasar**, **T.** (2026). Applications of engineering economy in industry. Invited panelist, *IISE Annual Conference & Expo*, Engineering Economy track. Arlington, TX, May 18.
4. **Cokyasar**, **T.** (2025). Designing a drone delivery network with automated battery swapping. Invited seminar, *Department of Mechanical and Industrial Engineering, University of Illinois Chicago*. Chicago, IL, October 10.
3. **Cokyasar**, **T.**, Verbas, O., & Bhuiyan, T. H. (2024). Optimizing e-commerce electric delivery vehicle charger locations under stochastic demand. *INFORMS Annual Meeting*. Invited abstract was presented in Seattle, WA, October 20-23.
2. **Cokyasar**, **T.** (2022). Last-mile drone and truck delivery energy comparisons. *INFORMS Annual Meeting*. Invited abstract was presented in Indianapolis, IN, October 16-19.
1. **Cokyasar**, **T.**, Dong, W., Jin, M., & Verbas, O. (2020). Designing a drone delivery network with automated battery swapping machines. *INFORMS Annual Meeting*. Abstract virtually presented as an invited speaker.

Conference Presentations

46. Acosta, J., Verbas, O., **Cokyasar**, **T.**, Auld, J., Altman, J., Beck, N., Alhajjar, M., Camargo, P., Cook, J., & Derrible, S. (2026). Agent-based modeling of transportation energy shifts and trends: A case study of the chicago region. *105th Transportation Research Board Annual Meeting*. Poster was jointly presented by Verbas, O. and Acosta, J. in Washington, DC, January 10-15.
45. Bassaleh, A., **Cokyasar**, **T.**, & Yildirim, M. B. (2026). Trajectory-based electric vehicle charging infrastructure planning *INFORMS Annual Meeting*. Abstract will be presented by Bassaleh, A. in San Francisco, CA, November 1-4.
44. Bazarnovi, S., **Cokyasar**, **T.**, Verbas, O., & Mohammadian, A. K. (2026). A unified framework for fleet composition, vehicle scheduling, and charging in electric diesel transit systems. *105th Transportation Research Board Annual Meeting*. Poster was presented in Washington, DC, January 10-15.
43. **Cokyasar**, **T.**, Bazarnovi, S., Verbas, O., Ismael, A., & Mohammadian, A. K. (2026). Optimizing bus fleet utilization through route interlining: A multi-agency case study. *105th Transportation Research Board Annual Meeting*. Poster was jointly presented by Verbas, O. and Cokyasar, T. in Washington, DC, January 10-15.
42. Hajibagher Najjar, Z., **Cokyasar**, **T.**, & Mohammadian, A. K. (2026). Multi-year fleet replacement and electrification planning for last-mile delivery with routing-based cost and mileage approximations. *INFORMS Annual Meeting*. Abstract will be presented by Najjar, Z. in San Francisco, CA, November 1-4.

41. Hosseini Dolatabadi, S. H., Bhuiyan, T. H., Kaleem, W., Subramanyam, A., & **Cokyasar**, **T.** (2026). A branch-and-cut algorithm for routing a heterogeneous drone-ground vehicle fleet to deliver time-sensitive products *INFORMS Annual Meeting*. Abstract will be presented by Hosseini Dolatabadi, H. in San Francisco, CA, November 1-4.
40. Ismael, A., & **Cokyasar**, **T.** (2026). From optimization to national scale: A two-stage framework for calibrated supplier selection in freight agent-based models. *INFORMS Annual Meeting*. Abstract will be presented by Ismael, A. in San Francisco, CA, November 1-4.
39. Ismael, A., & **Cokyasar**, **T.** (2026). Modeling and calibration of supplier selection problem in freight agent-based simulations. *105th Transportation Research Board Annual Meeting*. Study was presented in Washington, DC, January 10-15.
38. Mori, K., Gurumurthy, K. M., **Cokyasar**, **T.**, & Kockelman, K. (2026). Bayesian optimization of routes for a novel autonomous transit system. *INFORMS Annual Meeting*. Abstract will be presented by Mori, K. in San Francisco, CA, November 1-4.
37. Paul, J., **Cokyasar**, **T.**, Gurumurthy, K. M., Auld, J., & Jia, Y. (2026). Improving experience beyond the ride: Joint assignment and repositioning for rideshare. *105th Transportation Research Board Annual Meeting*. Poster was presented in Washington, DC, January 10-15.
36. Bazarnovi, S., **Cokyasar**, **T.**, Verbas, O., & Mohammadian, A. K. (2025). Strategic allocation of battery electric bus chargers under stochastic demand: A chicago case study using metaheuristic approaches. *104th Transportation Research Board Annual Meeting*. Poster was presented by Bazarnovi, S. in Washington, DC, January 5-9.
35. Borlaug, B., Sahin, O., Yip, A., Sun, J., **Cokyasar**, **T.**, Mansour, C., Jadun, P., & Muratori, M. (2025). Highly resolved transportation electricity load projections for the united states. *104th Transportation Research Board Annual Meeting*. Poster was presented by Borlaug, B. in Washington, DC, January 5-9.
34. Ismael, A., Sahin, O., Uhm, H.-s., Shen, H., Zuniga-Garcia, N., Huang, Y., **Cokyasar**, **T.**, Gurumurthy, K. M., Auld, J., & Stinson, M. (2025). Integration and application of a freight agent-based simulation framework. *104th Transportation Research Board Annual Meeting*. Poster was presented by Ismael, A. in Washington, DC, January 5-9.
33. Kaleem, W., **Cokyasar**, **T.**, Larson, J., Subramanyam, A., Verbas, O., & Bhuiyan, T. H. (2025). Large scale stochastic charger location and allocation problem in last-mile delivery . *104th Transportation Research Board Annual Meeting*. Poster was presented by Kaleem, W. in Washington, DC, January 5-9.
32. Ng, M., Tong, D., Mahmassani, H., Verbas, O., & **Cokyasar**, **T.** (2025). Joint optimization of pattern, headway, and fleet size of multiple urban transit lines with perceived headway consideration and passenger flow allocation. *104th Transportation Research Board Annual Meeting*. Poster was presented by Ng, M. in Washington, DC, January 5-9.
31. Paul, J., Su, H., Gurumurthy, K. M., **Cokyasar**, **T.**, Auld, J., & Jia, Y. (2025). A study of financial impacts of pooled rideshare based on assignment strategies. *104th Transportation Research Board Annual Meeting*. Poster was presented by Paul, J. in Washington, DC, January 5-9.
30. Sahin, O., Borlaug, B., **Cokyasar**, **T.**, Zuniga-Garcia, N., & Mansour, C. (2025). Analysis of load profiles for medium and heavy-duty electric vehicles: Implications for grid integration. *104th Transportation Research Board Annual Meeting*. Poster was presented by Sahin, O. in Washington, DC, January 5-9.
29. Verbas, O., **Cokyasar**, **T.**, Camargo, P. V. d., Gurumurthy, K. M., Zuniga-Garcia, N., & Auld, J. (2025). Modeling transit in a fully integrated agent-based framework: Methodology and large-scale application. *104th Transportation Research Board Annual Meeting*. Poster was presented by Verbas, O. in Washington, DC, January 5-9.
28. Verbas, O., **Cokyasar**, **T.**, Joyce-Johnson, S., Wainwright, S., Coates, M., Rousseau, A., Aloisi, J., Stewart, A., & Auld, J. (2025). Impact of transit on mobility, equity, and economy in the chicago metropolitan region. *104th Transportation Research Board Annual Meeting*. Poster was presented by Verbas, O. in Washington, DC, January 5-9.

27. Bazarnovi, S., **Cokyasar**, **T.**, Verbas, O., & Mohammadian, A. K. (2024). Locating and allocating battery electric bus chargers with stochastic charging demand. *INFORMS Annual Meeting*. Abstract was presented by Bazarnovi, S. in Seattle, WA, October 20-23.
26. Davatgari, A., **Cokyasar**, **T.**, Verbas, O., & Mohammadian, A. K. (2024). Heuristic approaches for the electric vehicle scheduling problem: Large-scale application with next day operability constraints. *103rd Transportation Research Board Annual Meeting*. Abstract was presented, Jan. 7-11.
25. Davatgari, A., Mohammadi, M. Y., **Cokyasar**, **T.**, & Mohammadian, A. K. (2024). A label-correcting algorithm for constrained one-to-many k-shortest path problem with replenishment. *103rd Transportation Research Board Annual Meeting*. Abstract was presented by Davatgari, A., Jan. 7-11.
24. Dean, M. D., Gurumurthy, K. M., Zhou, Z., Verbas, O., **Cokyasar**, **T.**, & Kockelman, K. (2024). An integrated transportation-power system model for a decarbonizing world. *103rd Transportation Research Board Annual Meeting*. Abstract was presented by Dean, M., Jan. 7-11.
23. Huang, Y., Verbas, O., Gurumurthy, K. M., Sahin, O., **Cokyasar**, **T.**, Zuniga, N., & Auld, J. (2024). Future transportation electrification: Operations and impacts . *Intelligent Transportation Systems America Conference and Expo*. Abstract was presented by Huang, Y. in Phoenix, AZ, May 25.
22. Ng, M., Mahmassani, H., Verbas, O., **Cokyasar**, **T.**, & Tong, D.. (2024). Train service design problem with perceived headwayconsideration and passenger flow allocation for urbanrail transit lines. *INFORMS Annual Meeting*. Abstract was presented in Seattle, WA, October 20-23.
21. Ng, M. T. M., Raymer, M., Mahmassani, H. S., Verbas, O., & **Cokyasar**, **T.** (2024). Can redesigning public transit network with shared autonomous mobility services improve equity? *103rd Transportation Research Board Annual Meeting*. Abstract was presented by Ng, M., Jan. 7-11.
20. Ng, M. T. M., Raymer, M., Mahmassani, H. S., Verbas, O., & **Cokyasar**, **T.** (2024). Spatial and vertical equity impacts of public transit network redesign with shared autonomous mobility services. *Transit Data 2024: The 9th International Workshop and Symposium on Research and Applications on the Use of Passive Data from Public Transport*. Abstract was presented by Ng, M. in London, UK, July 01-04.
19. Paul, J., Gurumurthy, K. M., **Cokyasar**, **T.**, Su, H., Jia, Y., & Auld, J. (2024). Integrating human factors in dynamic rideshare assignment: Willingness-to-pay for delay. *103rd Transportation Research Board Annual Meeting*. Abstract was presented by Paul, J., Jan. 7-11.
18. Verbas, O., & **Cokyasar**, **T.** (2024). Charger location and scheduling optimization for battery electric buses. *INFORMS International Meeting*. Abstract was presented by Verbas, O. in Medellin, Colombia, June 16-19.
17. Verbas, O., **Cokyasar**, **T.**, Rousseau, A., Auld, J., Joyce-Johnson, S., Aloisi, J., Stewart, A., & Barrett, G. (2024). Mobility, equity, and economic impact of transit in the Chicago metropolitan region. *INFORMS Annual Meeting*. Abstract will be presented in Seattle, WA, October 20-23.
16. **Cokyasar**, **T.**, Verbas, O., Davatgari, A., & Mohammadian, A. K. (2023). A heuristic solution to the single depot electric vehicle scheduling problem. *INFORMS Annual Meeting*. Abstract was presented by Verbas, O. in Phoenix, AZ, Oct. 15-18.
15. Davatgari, A., **Cokyasar**, **T.**, Subramanyam, A., Larson, J., & Mohammadian, A. K. (2023). Electric vehicle supply equipment allocation for fixed route vehicles. *IISE Annual Meeting & Expo*. Abstract was presented in New Orleans, LA, May 19-23.
14. Davatgari, A., **Cokyasar**, **T.**, Subramanyam, A., Larson, J., & Mohammadian, A. K. (2023). Locating and allocating chargers for fixed-route freight delivery. *INFORMS Annual Meeting*. Abstract was presented in Phoenix, AZ, Oct. 15-18.
13. Davatgari, A., **Cokyasar**, **T.**, Subramanyam, A., & Stinson, M. (2023). Electric vehicle supply equipment location and allocation for freight electrification. *Transportation Research Board Annual Meeting*. Poster presented by Davatgari, A. in Washington, DC, January 8-12.
12. Dean, M. D., Gurumurthy, K. M., Zhou, Z., Verbas, O., **Cokyasar**, **T.**, & Kockelman, K. (2023). An integrated transportation-power system model for a decarbonizing world. *Bridging Transportation Researchers (BTR) Online Conference*. Abstract was presented online by Dean, M., Aug. 9-10.

11. **Cokyasar**, **T.**, Larson, J., Stinson, M., Sahin, O., Verbas, O., Freyermuth, V., & Uhm, H. S. (2022). Truck routing optimization for large-scale transportation networks. *International Urban Freight Conference*. Abstract was presented by Uhm, H. S. in Long Beach, CA, May 25-27.
10. Subramanyam, A., **Cokyasar**, **T.**, Larson, J., & Stinson, M. (2022). Joint routing of conventional and range-extended electric vehicles in a large metropolitan network. *INFORMS Optimization Society Conference*. Abstract was presented by Subramanyam, A. in Greenville, SC, March 13-15.
9. Vakayil, A., Gurumurthy, K. M., de Souza, F., **Cokyasar**, **T.**, & Larson, J. (2022). Fast and scalable shared fleet algorithms for large-scale application. *INFORMS Annual Meeting*. Abstract was presented by Gurumurthy, K. M. in Indianapolis, IN, October 16-19.
8. Verbas, O., **Cokyasar**, **T.**, & Auld, J. (2022). Solving the electric vehicle scheduling problem at scale. *INFORMS Annual Meeting*. Abstract was presented by Verbas, O. in Indianapolis, IN, October 16-19.
7. **Cokyasar**, **T.** (2021). Planning routes for e-commerce delivery drones. *IISE Annual Meeting & Expo*. Abstract virtually presented.
6. Stinson, M., **Cokyasar**, **T.**, & Sahin, O. (2021). Mobility and sustainability impacts of advanced last- mile delivery technologies. *24th IEEE International Conference on Intelligent Transportation (ITSC2021) Last Mile Delivery and 50 Feet Logistics Challenges in the "New Normal" Workshop* Abstract was presented by Stinson, M. in Indianapolis, IN, September 19-22.
5. **Cokyasar**, **T.**, & Larson, J. (2020). Optimal assignment for the single-household shared autonomous vehicle problem. *IISE Annual Meeting & Expo*. Abstract virtually presented.
4. Verbas, O., **Cokyasar**, **T.**, & Auld, J. (2020). Chicago transit authority transit network efficiency and the changing mobility landscape. *U.S. Department of Energy's (DOE's) 2020 Annual Merit Review and Peer Evaluation Meeting (AMR) for the Hydrogen and Fuel Cells Program and the Vehicle Technologies Office, Poster Session EEMS088*. Poster virtually presented by Verbas, O.
3. **Cokyasar**, **T.**, Dong, W., & Jin, M. (2019). Network optimization for hybrid last mile delivery with trucks and drones. *INFORMS Computing Society*. Abstract was presented in Knoxville, TN.
2. **Cokyasar**, **T.**, Jin, M., & Willems, S. (2019). Spare parts supply chain optimization with additive manufacturing. *INFORMS Computing Society*. Abstract was presented in Knoxville, TN.
1. **Cokyasar**, **T.**, Jin, M., & Willems, S. (2018). Spare parts supply chain optimization with additive manufacturing. *INFORMS Annual Meeting*. Abstract was presented in Seattle, WA.

Teaching Experience

Texas A&M University, College Station, TX, U.S.A.

Fall 2026 • IDIS 450 Analytics for Distribution Operations, Sections 501/502 (52 students) – Department of Engineering Technology & Industrial Distribution

Fall 2026 • IDIS 450 Analytics for Distribution Operations, Sections 505/506 (45 students) – Department of Engineering Technology & Industrial Distribution

Tarsus University, Mersin, Turkey

Spring 2022 • IE 104 Computer Programming I – Department of Industrial Engineering

Spring 2022 • IE 532 Advanced Computer Programming – Department of Industrial Engineering

Fall 2021 • IE 530 Linear Programming and Network Optimization – Department of Industrial Engineering

Toros University, Mersin, Turkey

Fall 2021 • INE 321 Production Planning – Department of Industrial Engineering

The University of Tennessee, Knoxville, TN, U.S.A.

Fall 2019 • IE 430/530 Advanced/Supply Chain Engineering – Department of Industrial & Systems Engineering, *Teaching Score*: 3.9/5.0

Fall 2018 • IE 430/530 Advanced/Supply Chain Engineering – Department of Industrial & Systems Engineering, *Teaching Score*: 4.2/5.0

Advising and Mentorship

Current Graduate Students – Texas A&M University

2026 – Present • **Luke Bowie Kim**, Ph.D. Student, Industrial & Systems Engineering (My role: Chair).
Texas A&M University

Research Mentees and Co-Advised Students

2026 – Present • **Ahmad Bassaleh**, Research Collaborator (My role: Co-Mentor).
Wichita State University
Currently: Ph.D. Student.

2026 • **Yifan Wu**, Research Aide (My role: Primary Mentor).
Texas A&M University
Currently: Ph.D. Student.

2026 • **Annabelle Jung**, SULI Summer Intern (My role: Co-Mentor).
University of Illinois Chicago
Currently: B.S. Student.

2026 • **Waquar Kaleem**, Research Aide (My role: Primary Mentor).
Pennsylvania State University
Currently: Ph.D. Student.

2026 • **Hadi Bhidya**, SULI Spring Intern (My role: Primary Mentor).
University of Tennessee
Currently: B.S. Student.

2025 – Present • **Zahra Hajibagher Najjar**, Research Collaborator (My role: Primary Mentor).
University of Illinois Chicago
Currently: Ph.D. Student.

2024 – 2025 • **Draco Tong**, Research Collaborator (My role: Co-Mentor).
Northwestern University
Currently: Ph.D. Student.

2024 – 2025 • **Sadjad Bazarnovi, Ph.D.**, Research Collaborator (My role: Dissertation Co-Advisor).
University of Illinois Chicago.
Dissertation: *"Planning and Optimization of Charging Strategies for Electric Vehicles: From Infrastructure to Behavioral Insights."*
Currently: Operations Research Scientist at NetJets.

2024 • **Waquar Kaleem**, Givens Associate (My role: Co-Mentor).
Pennsylvania State University
Currently: Ph.D. Student.

2024 • **Selin Bayramoglu**, Research Aide (My role: Primary Mentor).
Georgia Institute of Technology.
Currently: Ph.D. Student.

2024 • **Joseph Paul, Ph.D.**, Research Aide (My role: Co-Mentor).
Clemson University.
Currently: Consultant at Plasman.

2023 – 2025 • **Max Ng, Ph.D.**, Research Collaborator (My role: Co-Mentor).
Northwestern University.
Currently: Machine Learning Engineer at Uber.

2023 • **Akhil Vakayil, Ph.D.**, Givens Associate (My role: Co-Mentor).
Georgia Institute of Technology.
Currently: Quantitative Analytics Specialist at Wells Fargo.

Advising and Mentorship (continued)

2022 – 2024 • **Amir Davatgari, Ph.D.**, Research Collaborator (My role: Dissertation Co-Advisor).
University of Illinois Chicago.
Dissertation: *"Medium-duty Electric Vehicle Infrastructure Planning and Operations in Urban Transportation Networks."*
Currently: Data Scientist at Starbucks.

Professional Service

Editorial Service

2025 – · · · · • *Area Editor, Infrastructure Economics* – The Engineering Economist
2021 – 2022 • *Editorial Board Member* – International Journal of Applied Management Science (IJAMS)

Society Leadership

2025 – 2027 • *President* – IISE Engineering Economy Division
2023 – 2025 • *Director* – IISE Logistics and Supply Chain Division
2022 – 2024 • *Director* – IISE Engineering Economy Division
2019 – 2021 • *President* – IISE Logistics and Supply Chain Division Student Leadership Board
2018 – 2019 • *Vice President* – IISE Logistics and Supply Chain Division Student Leadership Board

University and Department Service – Texas A&M University

2026 – 2027 • *Member* – ESET Faculty Search Committee (FY27), Department of Engineering Technology & Industrial Distribution
2026 – · · · · • *Member* – Academic Professional Track Faculty Search Committee (Industrial Distribution, open rank), Department of Engineering Technology & Industrial Distribution

Conference Organization

2026 • *Session Chair* – 2026 INFORMS Annual Meeting (Transportation Science & Logistics Society)
2026 • *Session Chair* – 2026 IISE Annual Conference & Expo, "Transportation and Infrastructure Optimization" (Engineering Economy track)
2026 • *Track Chair* – 2026 IISE Annual Meeting & Expo (Engineering Economy); the 17th International Conference on Emerging Ubiquitous Systems and Pervasive Networks (EUSPN – Adaptive Systems, Intelligent Computing and Applications)
2025 • *Track Chair* – 2025 IISE Annual Meeting & Expo (Engineering Economy & Facilities Design and Planning); the 16th International Conference on Emerging Ubiquitous Systems and Pervasive Networks (EUSPN – Adaptive Systems, Intelligent Computing and Applications)
2024 • *Track Chair* – 2024 IISE Annual Meeting & Expo (Engineering Economy); the 21st International Conference on Mobile Systems and Pervasive Computing (Mobi-SPC – Smart Cities and Ubiquitous Climate Change Management)
2023 • *Track Chair* – 2023 IISE Annual Meeting & Expo (Engineering Economy); the 14th International Conference on Emerging Ubiquitous Systems and Pervasive Networks (EUSPN – Adaptive Systems, Intelligent Computing and Applications)

Peer Review Activities

2016 – · · · · • *Ad-hoc Reviewer* (78 reviews, 2020–2026) – Transportation Science, Omega, European Journal of Operational Research (EJOR), Transportation Research Part A: Policy and Practice (TRA), Transportation Research Part B: Methodological, Transportation Research Part C: Emerging Technologies (TRC), Transportation Research Part E: Logistics and Transportation Review (TRE), IISE Annual Meeting & Expo, IEEE Transactions on Engineering Management (TEM), IEEE Intelligent Transportation Systems Conference (ITSC), Computers & Operations Research (COR), Journal of Cleaner Production (JCLP), Transportation Research Board (TRB), Transportation Research Record (TRR), International Journal of Simulation and Process Modelling (IJSPM), Transportation Letters

Professional Service (continued)

Professional Memberships

2018 – · · · · • *Member* – INFORMS
2016 – · · · · • *Member* – IISE

Honors and Awards

Professional & Academic Honors

2024 • *Impact Argonne Award (Extraordinary Effort)* – Argonne National Laboratory
2018 • *Best Track Paper Award* – Logistics & Supply Chain Division of IISE for the conference paper: "An Agent-Based Simulation Model for Optimization of Spare Parts Sourcing with Additive Manufacturing" by Cokyasar T., Kose H., Dong W., Li X., Jin M., and Li W.
2018 • *2nd Best Track Paper Award* – Engineering Management Division of IISE for the conference paper: "Optimization of Maintenance and Infrastructure Projects Based on Risk Factors" by Hassler M., Cokyasar T., Jin M., and Tang R.
2018 • *Extraordinary Professional Promise Award* – Chancellor's Citation Awards, the University of Tennessee
2017 • *Service Recognition Award* – Student Disability Services, the University of Tennessee

Fellowships & Scholarships

2019 – 2020 • *E.J. Sierleja Memorial Fellowship* – IISE
2017 – 2019 • *Helen Jubin Fellowship Award* – College of Engineering, the University of Tennessee
2013 – 2020 • *Graduate Fellowship Award* – Republic of Türkiye Ministry of National Education
2008 – 2012 • *Undergraduate Fellowship Award* – Renaissance Education Foundation (Rönesans Eğitim Vakfı)

Other Experience

12/2013 – 05/2020 • Government Sponsored Scholar (YLSY program) – *Republic of Türkiye Ministry of National Education*, Ankara, Türkiye
08/2015 – 03/2017 • Note Taker & Test Proctor, Student Disability Services – *The University of Tennessee*, Knoxville, TN, U.S.A.
08/2012 – 07/2014 • Retail and Small Business Sales Assistant – *Kuwait-Turkish Participation Bank*, Adana, Türkiye
06/2011 – 08/2011 • Operations and Accounting Intern – *Vakifbank*, Adana, Türkiye
06/2009 – 09/2009 • Sales Representative – *Six Flags Darien Lake*, Buffalo, NY & *Great Adventure*, Jackson, NJ, U.S.A.

Technical Skills

Programming • Python (pandas, NumPy, GeoPandas), R, Java, MATLAB, SQL
Optimization • Gurobi, CPLEX, PYOMO, AMPL, KNITRO, GLPK, IPOPT
Software & Tools • QGIS, Anylogic, SPSS, Git/GitHub, LaTeX
Databases • HDF5, SQLite
Languages • Turkish (Native), English (Fluent)